

50.007 Machine Learning

Hidden Markov Model

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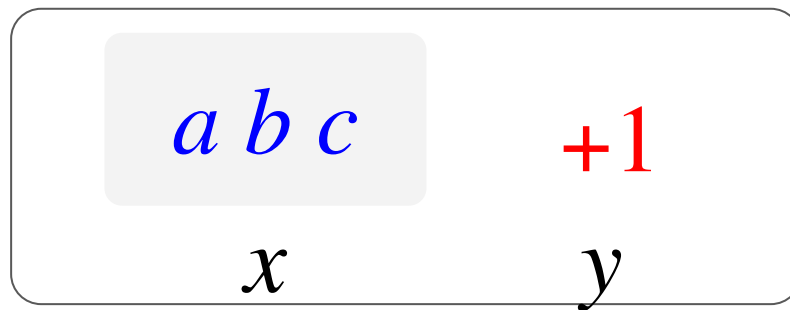


Learning Objectives

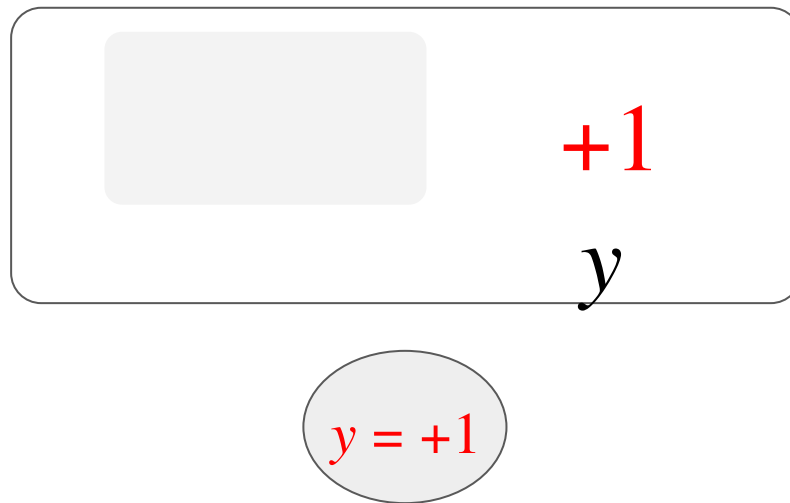
You should be able to know:

- What is a **Hidden Markov Model (HMM)**? What **types of data** it can handle?
- What is the **relation** between **HMM** and **mixture model**?
- How to **describe the joint distribution** of a HMM with the **transition** and **emission probabilities**?
- How to **learn parameters** of HMM in **supervised** learning setting?

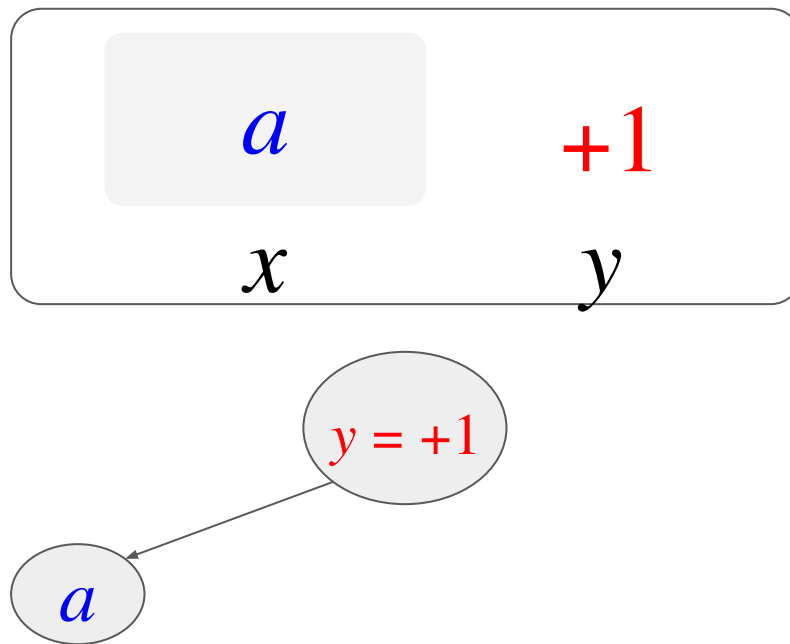
Generative Models



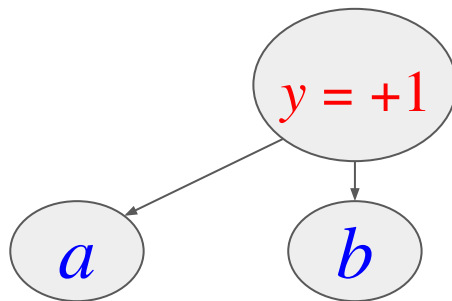
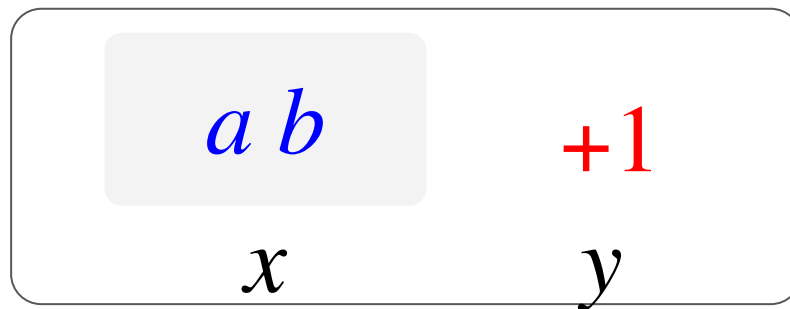
Generative Models



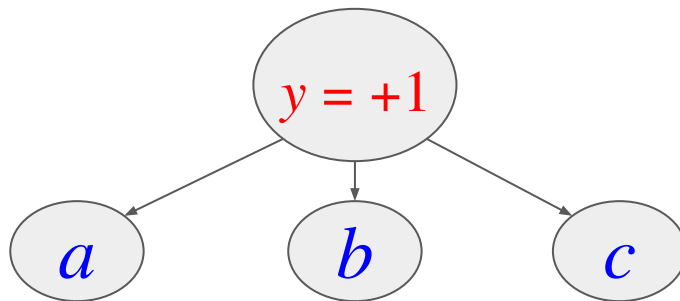
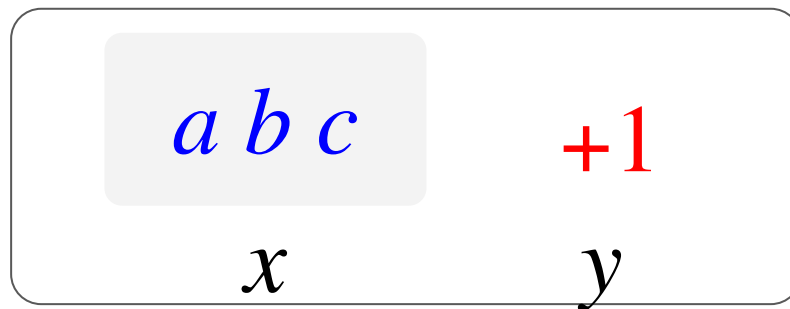
Generative Models



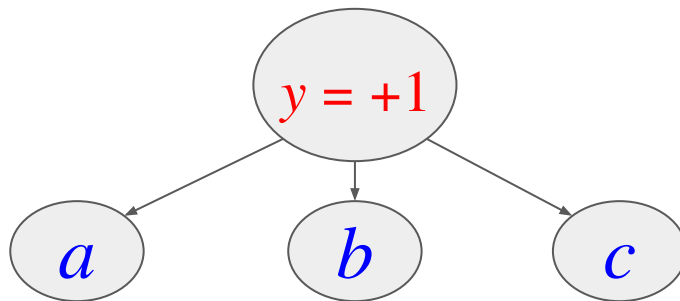
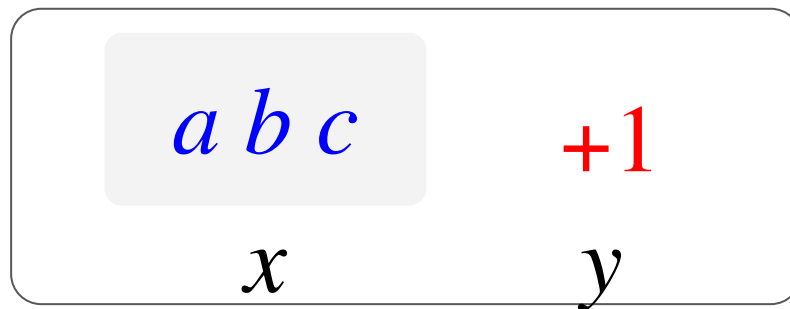
Generative Models



Generative Models

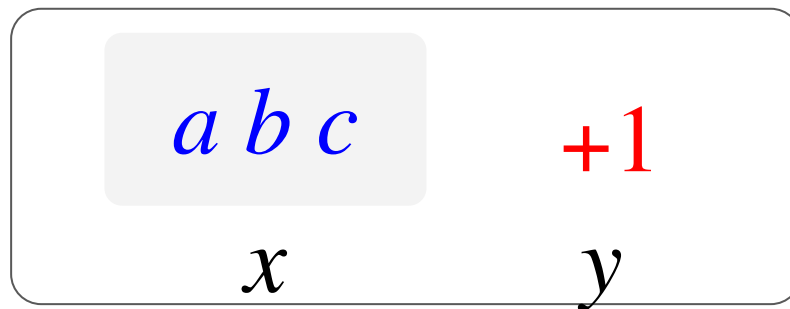


Generative Models

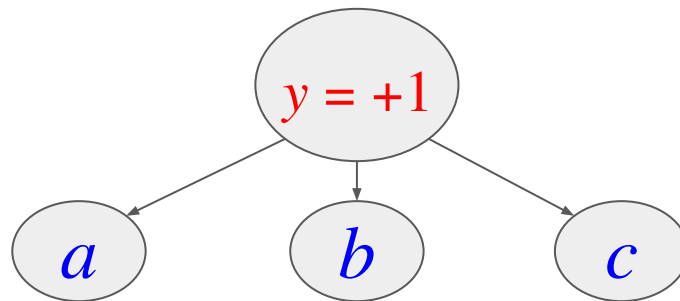


$$p(x = \text{"a, b, c"}, y = +1)$$

Generative Models



Joint model.



Gaussian model



$$p(x = "a, b, c", y = +1) = p(y = +1) p(a | y = +1) p(b | y = +1) p(c | y = +1)$$

Sequence Labeling

Faith is a fine invention

Sequence Labeling

	Noun	Verb	Determiner	Adjective	Noun
✓ →	N	V	D	A	N
✗ →	Faith	is	a	fine	invention

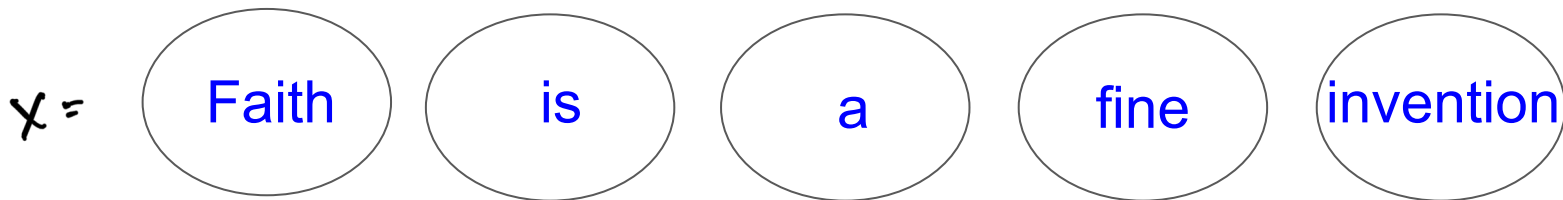
Sequence Labeling



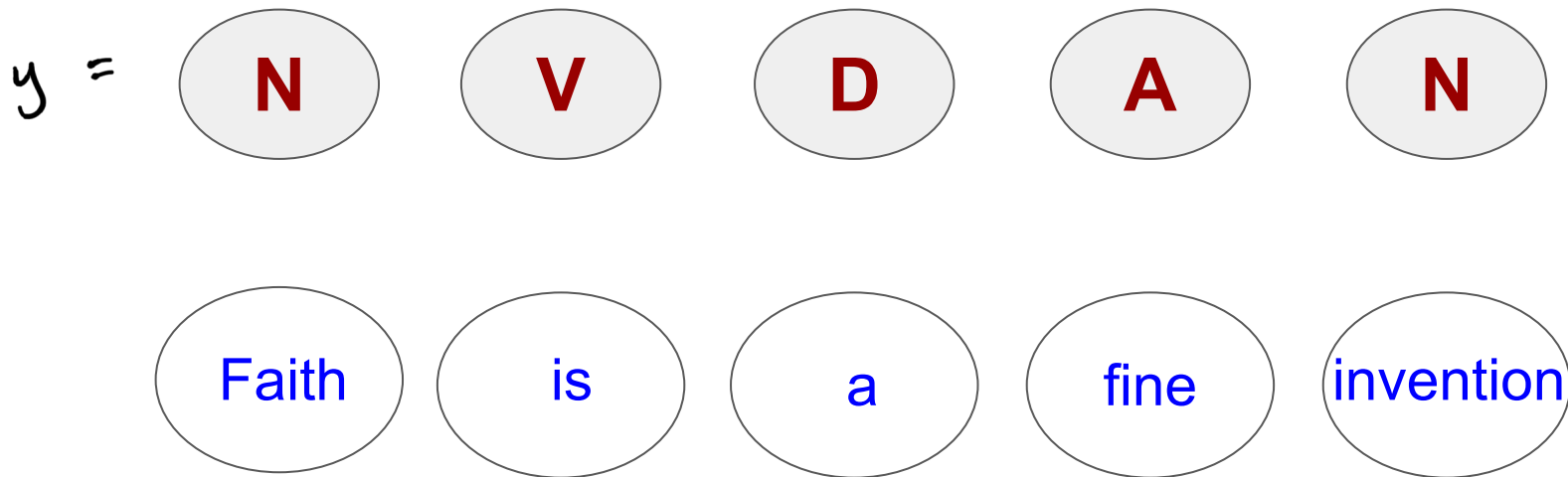
Sequence Labeling



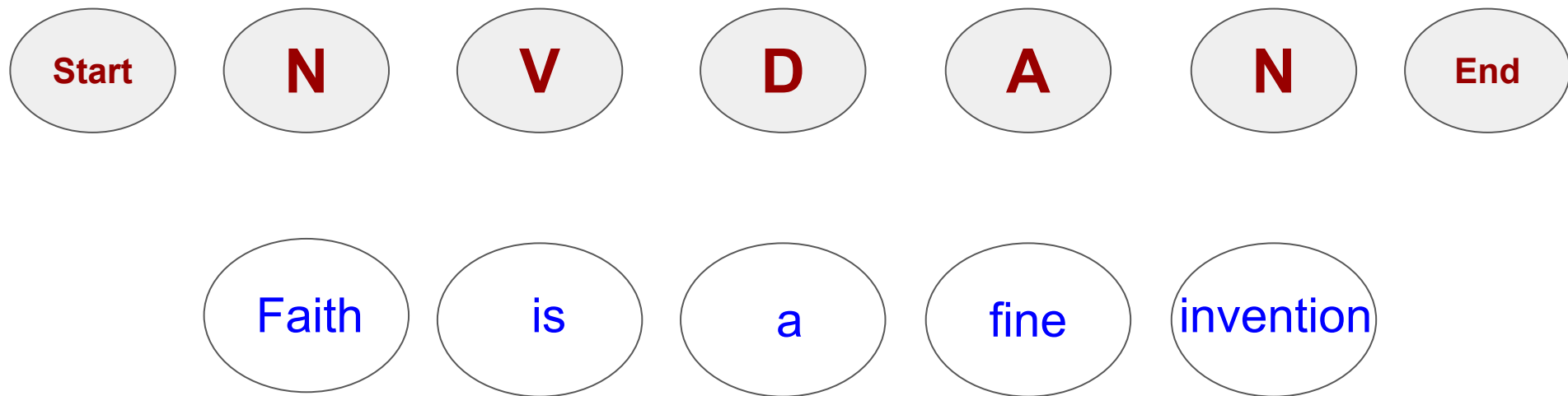
Sequence Labeling



Sequence Labeling



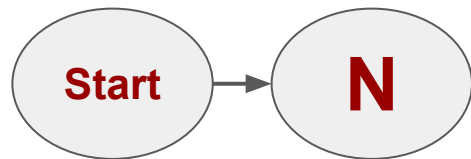
Sequence Labeling



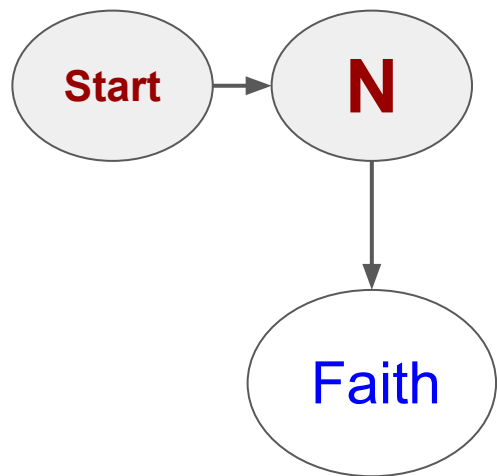
Sequence Labeling

Start

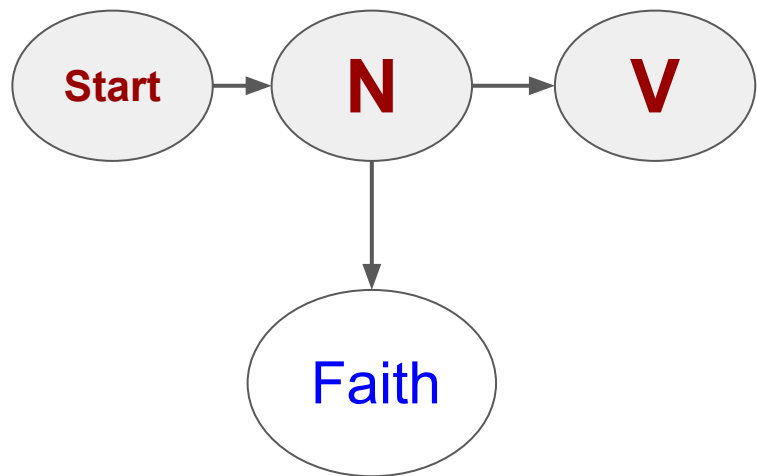
Sequence Labeling



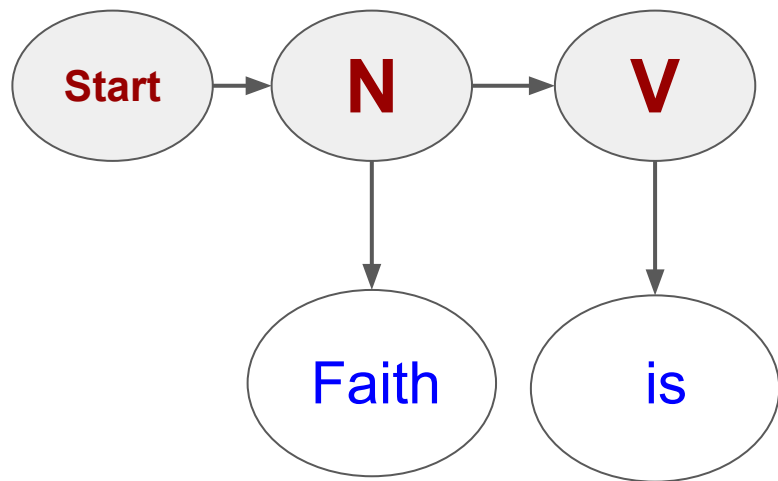
Sequence Labeling



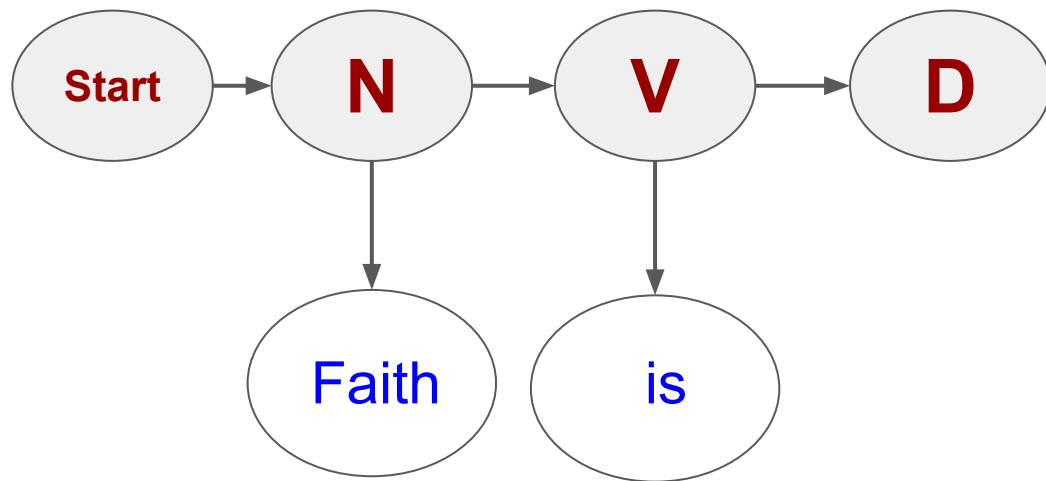
Sequence Labeling



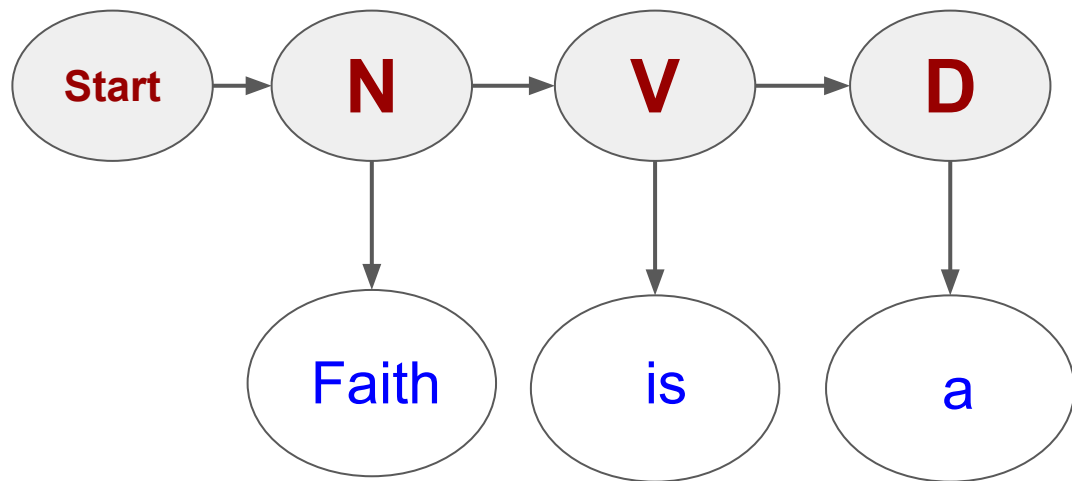
Sequence Labeling



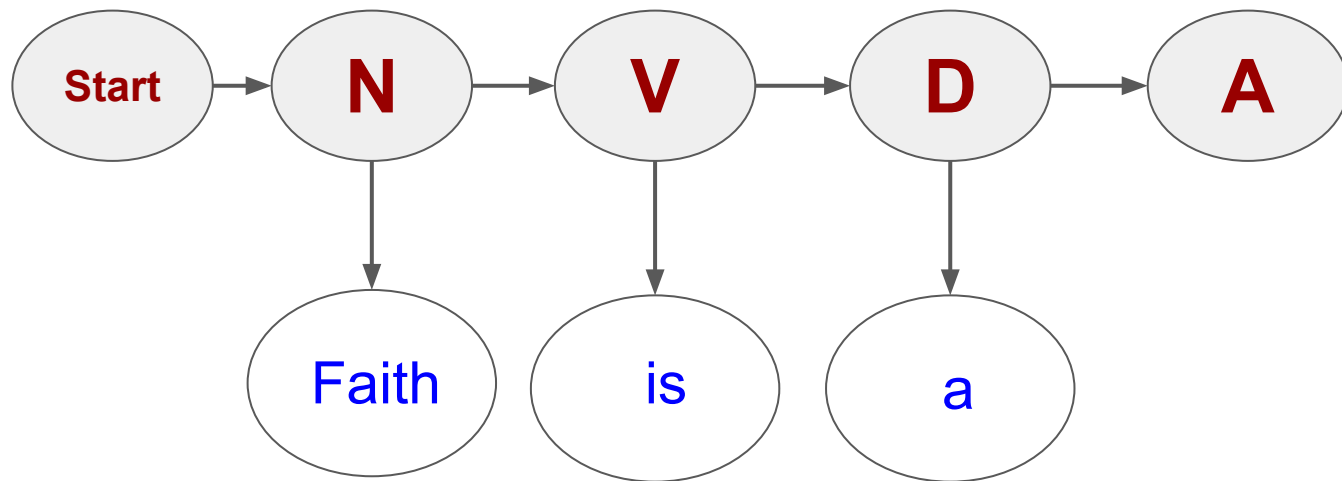
Sequence Labeling



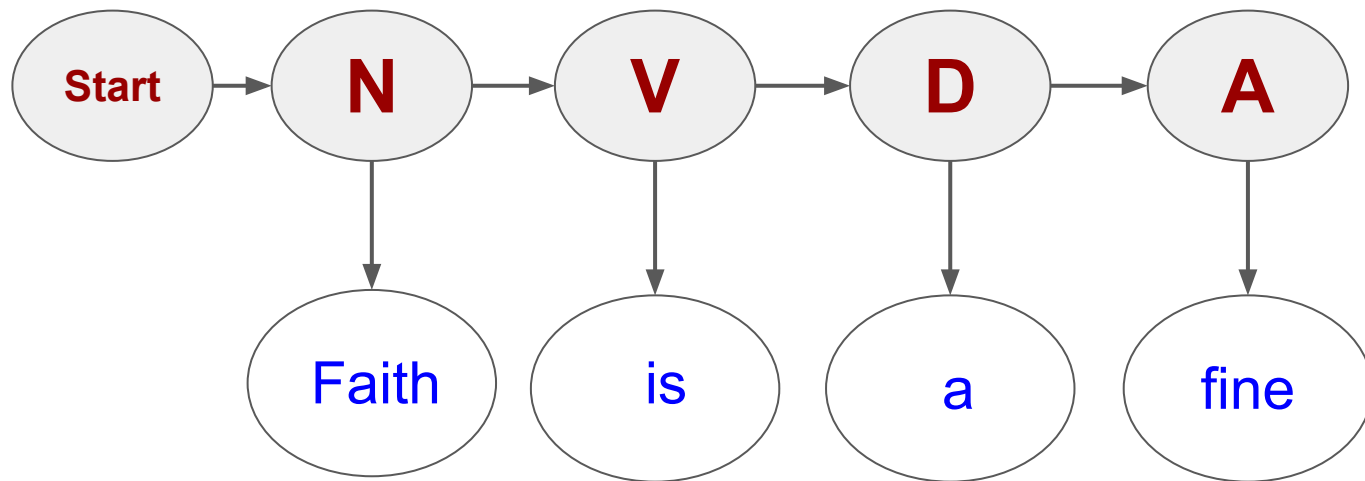
Sequence Labeling



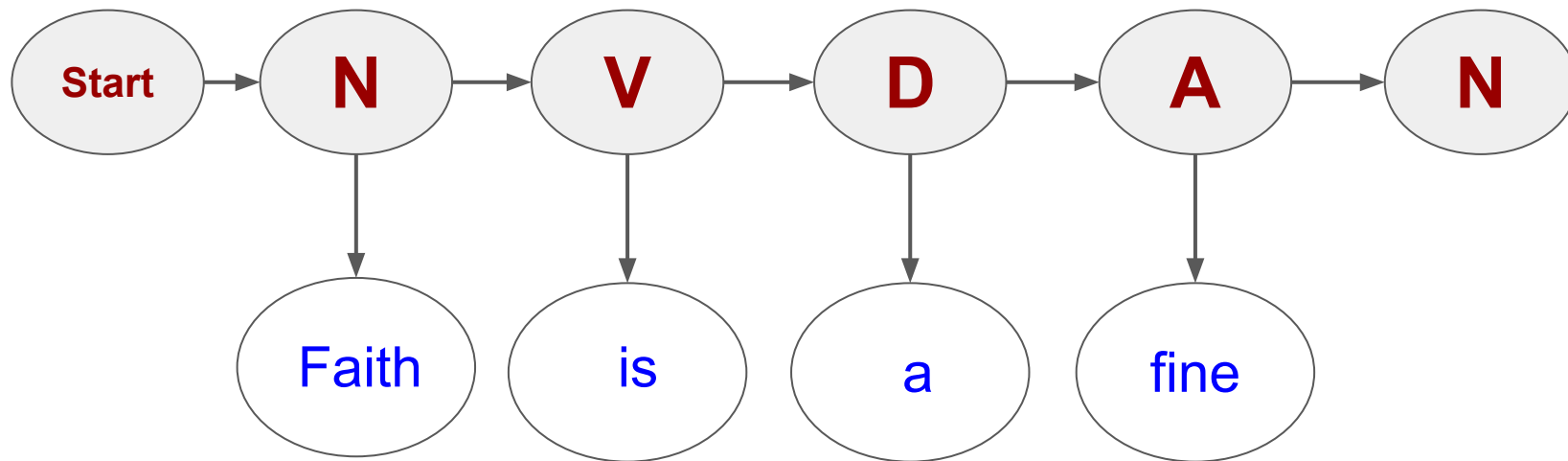
Sequence Labeling



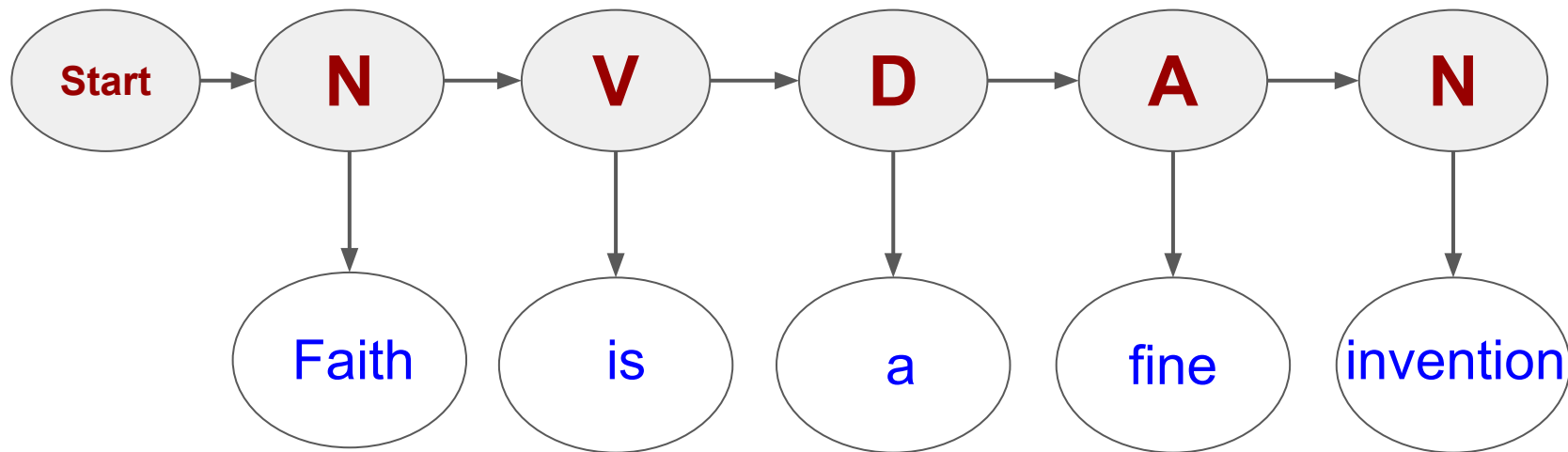
Sequence Labeling



Sequence Labeling

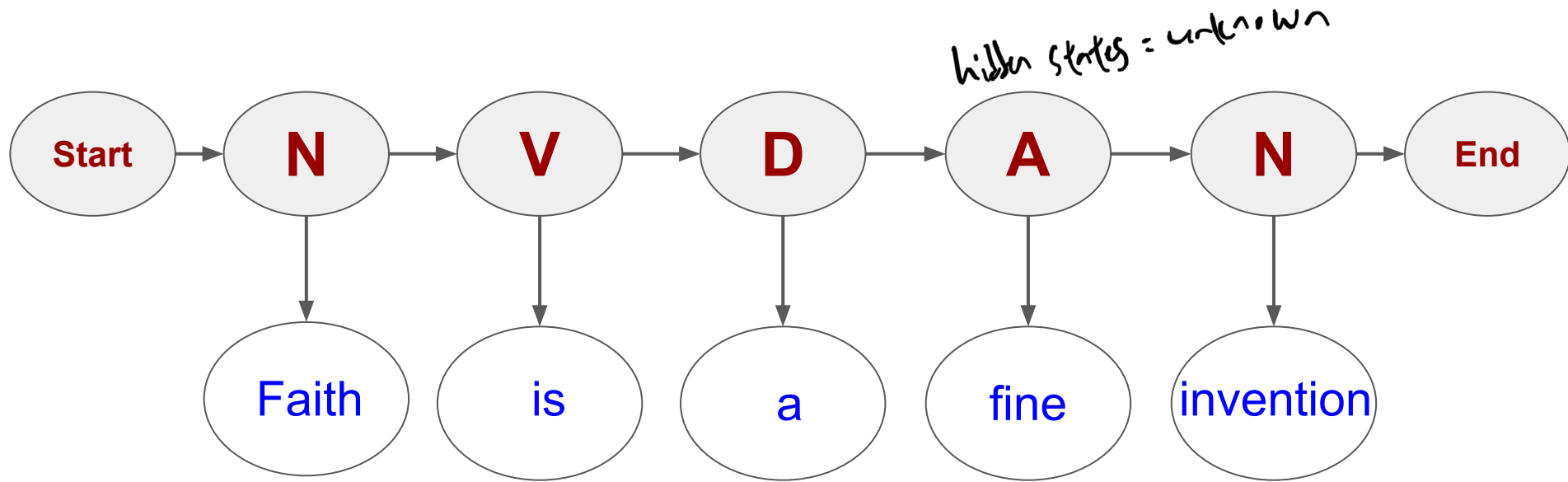


Sequence Labeling

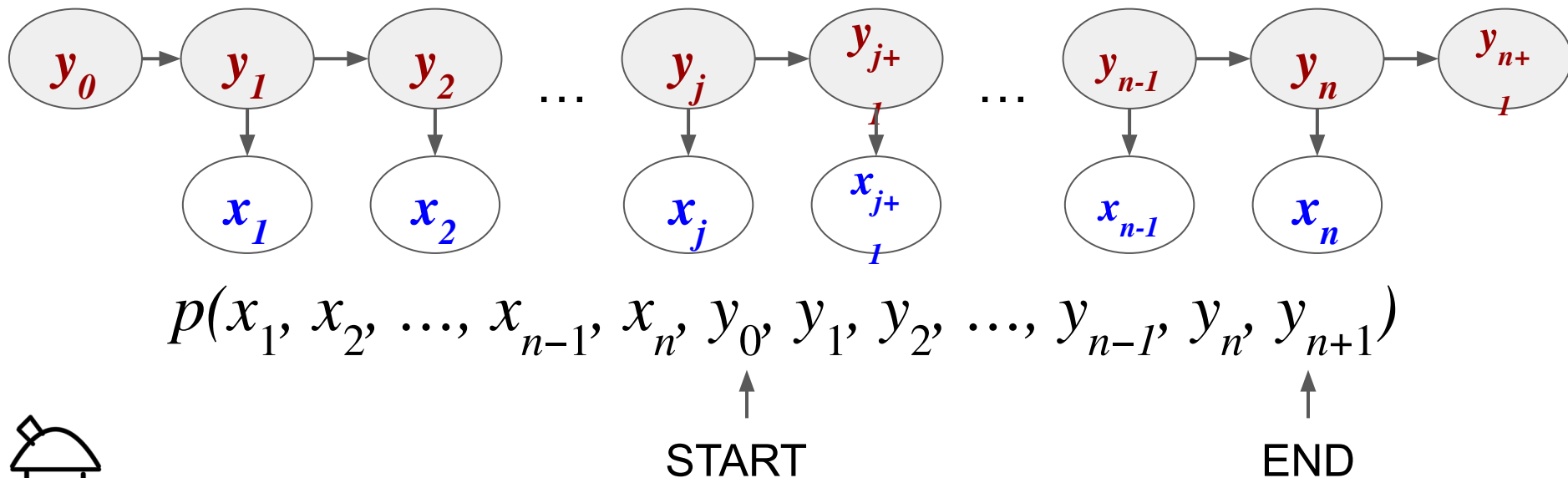


Sequence Labeling

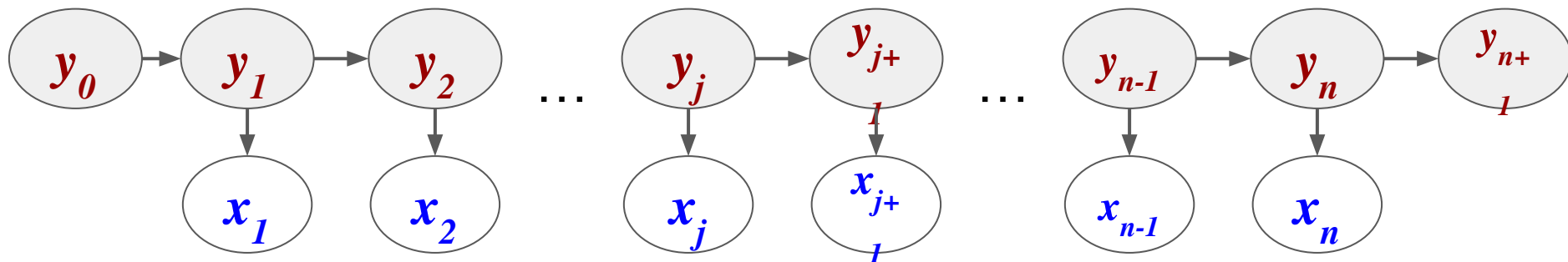
y is hidden $\Rightarrow$ state region



Sequence Labeling



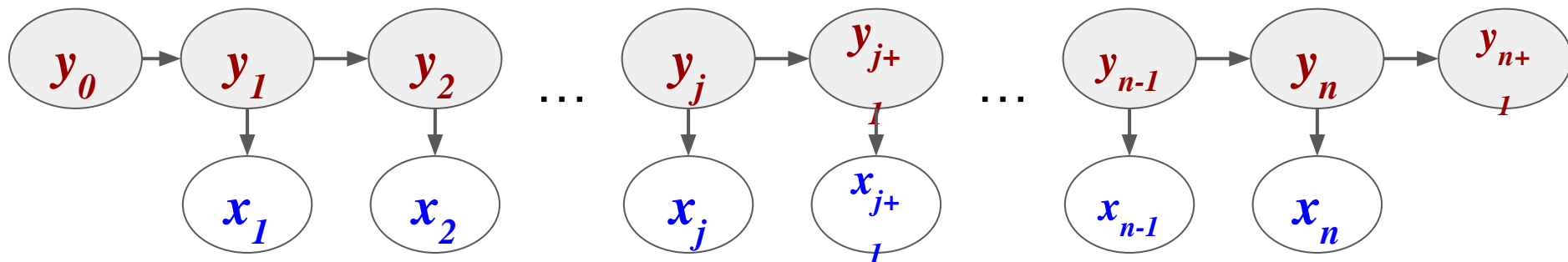
Sequence Labeling



$$p(x_1, x_2, \dots, x_{n-1}, x_n, y_0, y_1, y_2, \dots, y_{n-1}, y_n, y_{n+1})$$

$$\prod_{j=0}^n p(y_{j+1}|y_j) \times \prod_{j=1}^n p(x_j|y_j)$$

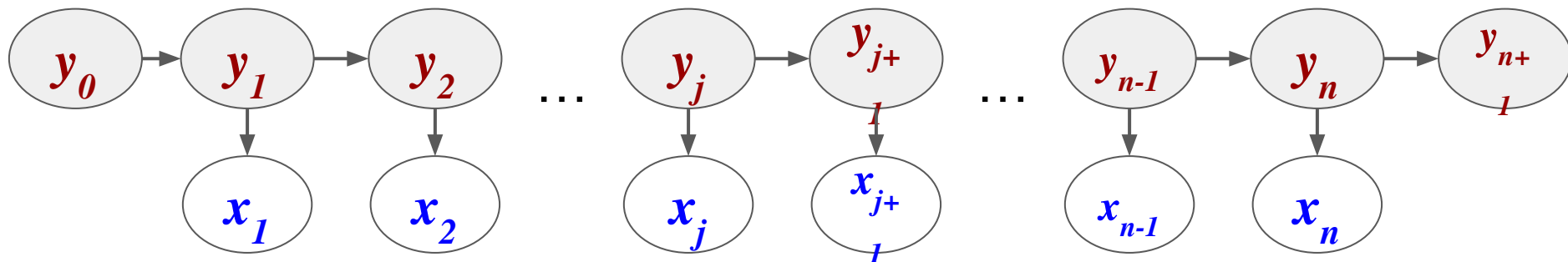
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$$p(x_1, x_2, \dots, x_{n-1}, x_n, y_0, y_1, y_2, \dots, y_{n-1}, y_n, y_{n+1})$$

$$\prod_{j=0}^n p(y_{j+1}|y_j) \times \prod_{j=1}^n p(x_j|y_j)$$

Sequence Labeling



$$p(x_1, x_2, \dots, x_{n-1}, x_n, y_0, y_1, y_2, \dots, y_{n-1}, y_n, y_{n+1})$$

can be represented $\leftarrow$ $\prod_{j=0}^n a_{y_j, y_{j+1}}$ $\times$ $\prod_{j=1}^n b_{y_j}(x_j)$

in a matrix $\Rightarrow$ α matrix / transition matrix

Transition Probabilities

Emission Probabilities

Hidden Markov Model

- An HMM is defined by a tuple $\langle \mathcal{T}, \mathcal{O}, \theta \rangle$, where
 - $\mathcal{T}$: a set of states including START and END states
 - $\mathcal{O}$: a set of observation symbols
 - θ : Transition and emission parameters $a_{u,v}$, and $b_u(o)$

Hidden Markov Model (Example)

$$\mathcal{T} = \{\text{START}, A, B, \text{STOP}\}$$

$$\mathcal{O} = \{\text{"the"}, \text{"dog"}\}$$

$u \backslash v$	A	B	STOP
START	1.0	0.0	0.0
A	0.5	0.5	0.0
B	0.0	0.8	0.2

$$a_{u,v}$$

$u \backslash o$	"the"	"dog"
A	0.9	0.1
B	0.1	0.9

$$b_u(o)$$

Hidden Markov Model (Example)

$a_{u,v}$

$u \backslash v$	A	B	STOP
START	1.0	0.0	0.0
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B	0.0	0.8	0.2

$b_u(o)$

$u \backslash o$	"the"	"dog"
A	0.9	0.1
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$(x, y) = \text{the}/A, \text{dog}/B, \text{the}/A$

What is P of
the, dog, the
given A, B, A

What is $p(x, y)$?

$P(x = \text{ } | y = \text{ })$?



Hidden Markov Model (Example)

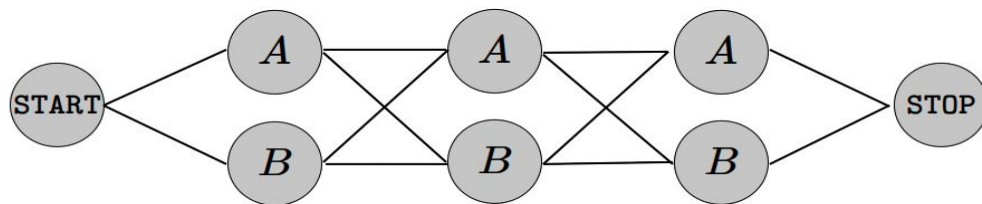
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$(\mathbf{x}, \mathbf{y}) = \text{the}/A, \text{dog}/B, \text{the}/A$



Hidden Markov Model (Example)

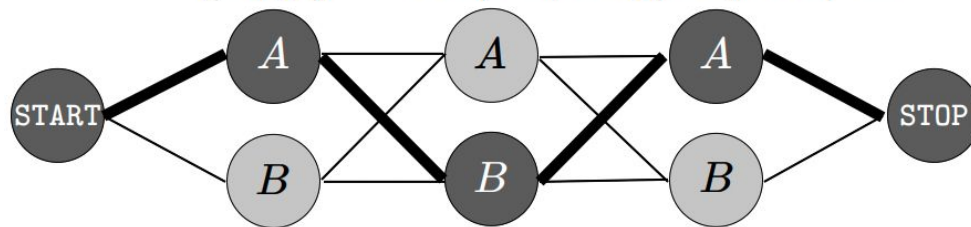
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$(\mathbf{x}, \mathbf{y}) = \text{the}/A, \text{dog}/B, \text{the}/A$



$a_{\text{START},A} = 1.0$

Hidden Markov Model (Example)

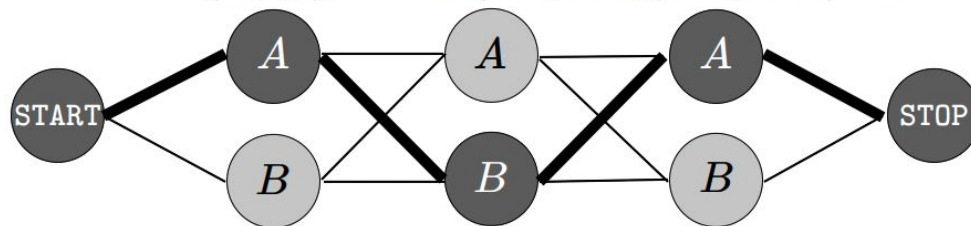
$a_{u,v}$

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A	0.9	0.1
B	0.1	0.9

$(\mathbf{x}, \mathbf{y}) = \text{the}/A, \text{dog}/B, \text{the}/A$



$$a_{\text{START}, A} \times b_A(\text{"the"}) = 1.0 \times 0.9$$

Hidden Markov Model (Example)

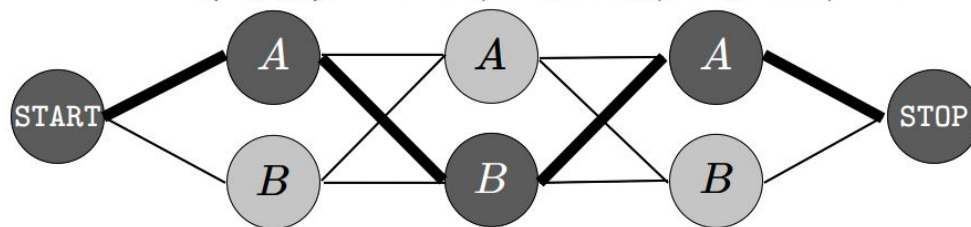
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A	0.9	0.1
B	0.1	0.9

$(\mathbf{x}, \mathbf{y}) = \text{the}/A, \text{dog}/B, \text{the}/A$



$$a_{\text{START},A} \times b_A(\text{"the"}) \times a_{A,B} = 1.0 \times 0.9 \times 0.5$$

Hidden Markov Model (Example)

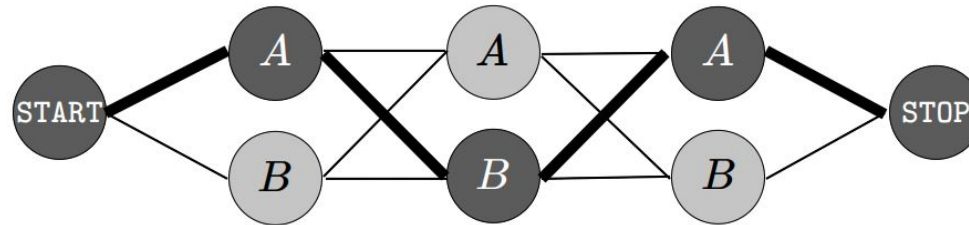
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$(\mathbf{x}, \mathbf{y}) = \text{the}/A, \text{dog}/B, \text{the}/A$



$$a_{\text{START},A} \times b_A(\text{"the"}) \times a_{A,B} \times b_B(\text{"dog"}) = 1.0 \times 0.9 \times 0.5 \times 0.9$$

Hidden Markov Model (Example)

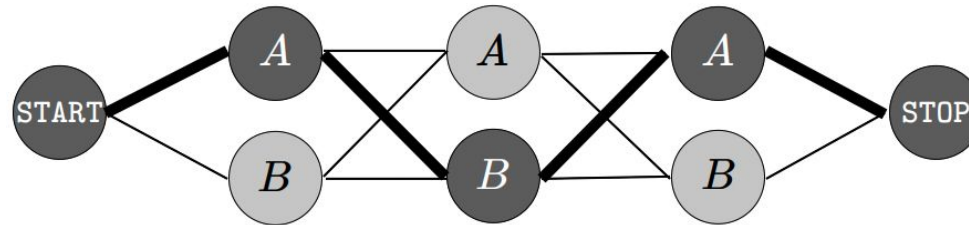
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$$a_{\text{START},A} \times b_A(\text{"the"}) \times a_{A,B} \times b_B(\text{"dog"}) \times a_{B,A} = 1.0 \times 0.9 \times 0.5 \times 0.1 \times 0.0$$

Hidden Markov Model (Example)

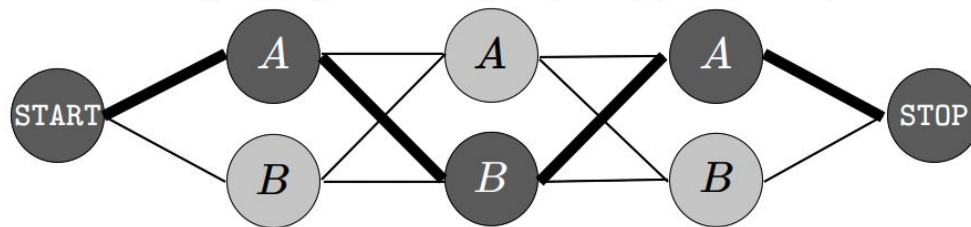
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$(\mathbf{x}, \mathbf{y}) = \text{the}/A, \text{dog}/B, \text{the}/A$



$$a_{\text{START},A} \times b_A(\text{"the"}) \times a_{A,B} \times b_B(\text{"dog"}) \times a_{B,A} \times b_A(\text{"the"})$$

$$1.0 \times 0.9 \times 0.5 \times 0.9 \times 0.0 \times 0.1$$

$$= 0.9$$

Hidden Markov Model (Example)

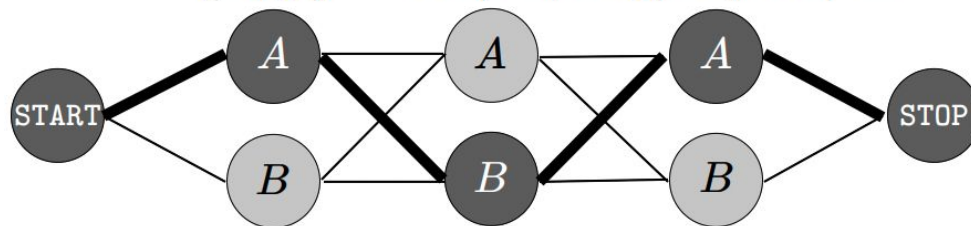
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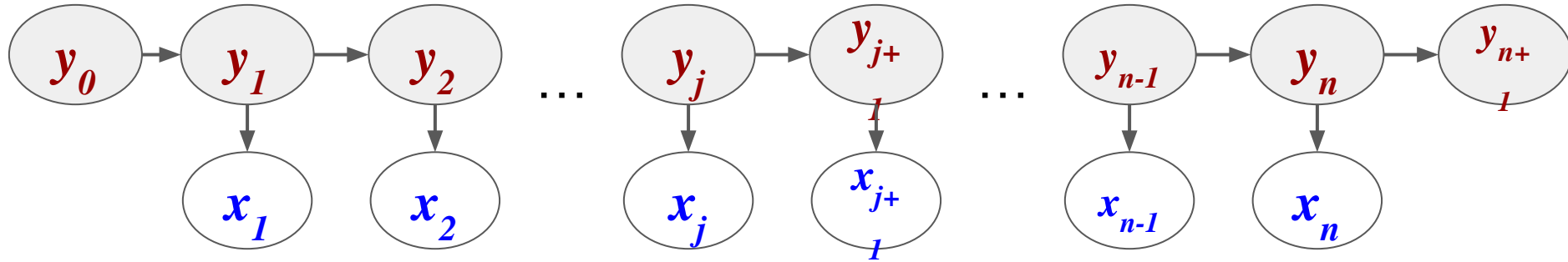
$(\mathbf{x}, \mathbf{y}) = \text{the}/A, \text{dog}/B, \text{the}/A$



$$1.0 \times 0.1 \times 0.5 \times 0.1 \times 0.0 \times 0.1 \times 0.0$$

$$a_{\text{START},A} \times b_A(\text{"the"}) \times a_{A,B} \times b_B(\text{"dog"}) \times a_{B,A} \times b_A(\text{"the"}) \times a_{A,\text{STOP}} = 0.0$$

HMM - Supervised Learning

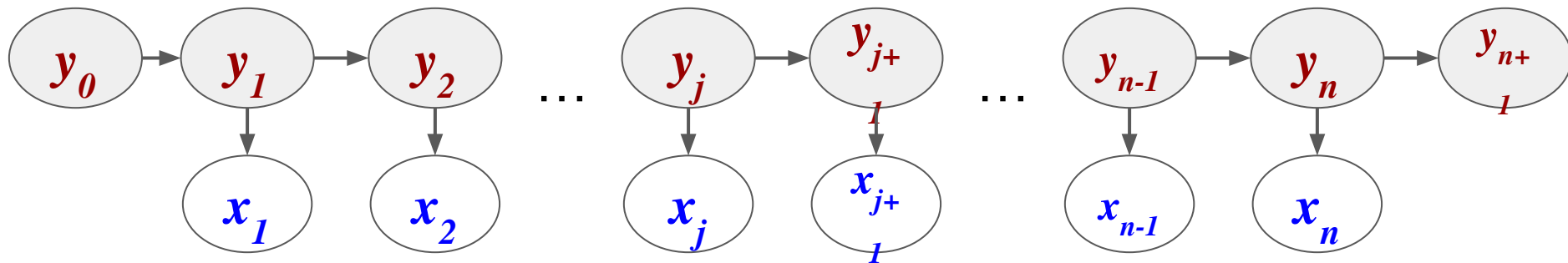


$$p(\mathbf{x}, \mathbf{y}) = \underbrace{\prod_{j=0}^n a_{y_j, y_{j+1}}}_{\text{Transition probabilities}} \times \underbrace{\prod_{j=1}^n b_{y_j}(x_j)}_{\text{Emission probabilities}}$$

How do we get the model parameters? (a and b matrix)? I.e, how do we do the learning?

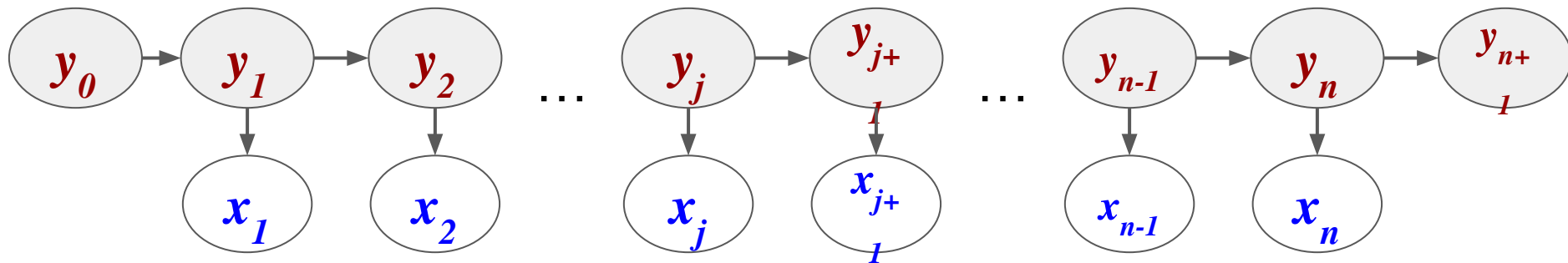


HMM - Supervised Learning



$$a_{u,v} = \frac{\text{count}(u,v)}{\text{count}(u)} \quad b_u(o) = \frac{\text{count}(u \rightarrow o)}{\text{count}(u)}$$

HMM - Supervised Learning



Number of times we see a transition from u to v

$$a_{u,v} = \frac{\text{count}(u,v)}{\text{count}(u)}$$

Number of times we see the state u in the training set

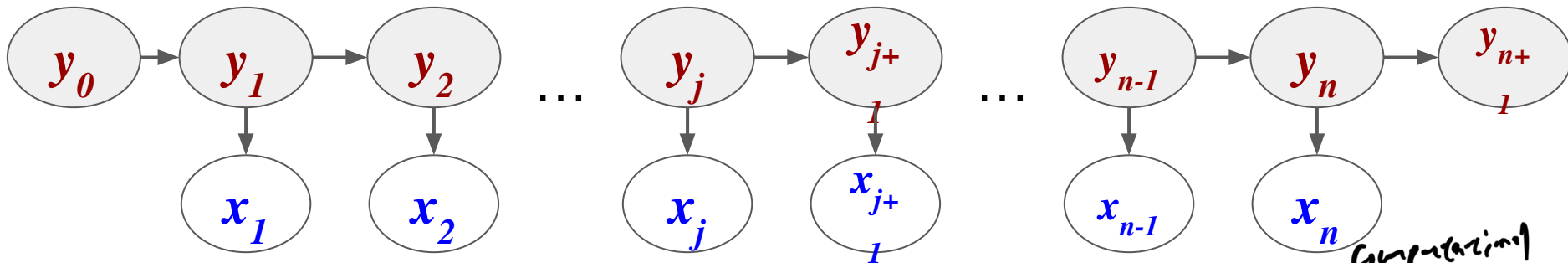
Number of times we see observation o generated from u

$$b_u(o) = \frac{\text{count}(u \rightarrow o)}{\text{count}(u)}$$

Number of times we see the state u in the training set



HMM - Supervised Learning



Number of times we see a transition from u to v

$$a_{u,v} = \frac{\text{count}(u,v)}{\text{count}(u)}$$

Number of times we see the state u in the training set

Number of times we see observation o generated from u

$$b_u(o) = \frac{\text{count}(u \rightarrow o)}{\text{count}(u)}$$

Number of times we see the state u in the training set

Still quite tough... Got better way to **estimate** the parameters?

Computationally invisible "take too much time."



HMM - Supervised Learning

We estimate the parameters of the HMM by performing **Maximum Likelihood Estimation**.

$$p(\mathbf{x}, \mathbf{y}) = \underbrace{\prod_{j=0}^n a_{y_j, y_{j+1}}}_{\text{Transition probabilities}} \times \underbrace{\prod_{j=1}^n b_{y_j}(x_j)}_{\text{Emission probabilities}}$$

By setting the derivative of log-likelihood *w.r.t.* each individual parameter to zero, we can get a close-form estimate for each parameter:

Number of times we see a transition from u to v

$$a_{u,v} = \frac{\text{count}(u,v)}{\text{count}(u)}$$

Number of times we see the state u in the training set

Number of times we see observation o generated from u

$$b_u(o) = \frac{\text{count}(u \rightarrow o)}{\text{count}(u)}$$

Number of times we see the state u in the training set



1 following slides

HMM - Supervised Learning (Example)

Let us use a simple labeled sequence dataset to help you understand how we estimate the parameters in the supervised learning setting.

- The labeled training dataset contains three sequence pairs:

$y^{(1)} = (A, B)$	$y^{(2)} = (A, B)$	$y^{(3)} = (A, B)$
$x^{(1)} = (e, g)$	$x^{(2)} = (e, h)$	$x^{(3)} = (f, h)$

$O = \{e, f, g, h\}$, and $T = \{START, A, B, STOP\}$

$$O = \{e, f, g, h\} \quad T = \{START, A, B, STOP\}$$

HMM - Supervised Learning (Example)

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$O = \{e, f, g, h\}$, and $I = \{START, A, B, STOP\}$

- 1) We first write down the likelihood:

$$\rightarrow a_{START \rightarrow A} \times b_A(e) \times a_{A \rightarrow B} \times b_B(g) \times a_{B \rightarrow STOP}$$

$$a_{START, A} \times b_A(e) \times a_{A, B} \times b_B(g) \times a_{B, STOP}$$

$$\times a_{START, A} \times b_A(e) \times a_{A, B} \times b_B(h) \times a_{B, STOP}$$

$$\times a_{START, A} \times b_A(f) \times a_{A, B} \times b_B(h) \times a_{B, STOP}$$

HMM - Supervised Learning (Example)

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- The labeled training dataset contains three sequence pairs:

$$\begin{array}{ccc} y^{(1)} = (A, B) & y^{(2)} = (A, B) & y^{(3)} = (A, B) \\ x^{(1)} = (e, g) & x^{(2)} = (e, h) & x^{(3)} = (f, h) \end{array}$$

$O = \{e, f, g, h\}$, and $I = \{START, A, B, STOP\}$

- 1) We first write down the likelihood:

$$\begin{aligned} & a_{START,A} \times b_A(e) \times a_{A,B} \times b_B(g) \times a_{B,STOP} \\ & \times a_{START,A} \times b_A(e) \times a_{A,B} \times b_B(h) \times a_{B,STOP} \\ & \times a_{START,A} \times b_A(f) \times a_{A,B} \times b_B(h) \times a_{B,STOP} \end{aligned}$$

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$$\begin{array}{ccc} y^{(1)} = (A, B) & y^{(2)} = (A, B) & y^{(3)} = (A, B) \\ x^{(1)} = (e, g) & x^{(2)} = (e, h) & x^{(3)} = (f, h) \end{array}$$

$O = \{e, f, g, h\}$, and $I = \{START, A, B, STOP\}$

- 1) We first write down the likelihood:

$$\begin{aligned} & a_{START,A} \times b_A(e) \times a_{A,B} \times b_B(g) \times a_{B,STOP} \\ & \times a_{START,A} \times b_A(e) \times a_{A,B} \times b_B(h) \times a_{B,STOP} \\ & \times a_{START,A} \times b_A(f) \times a_{A,B} \times b_B(h) \times a_{B,STOP} \\ & = (a_{START,A})^3 \times (a_{A,B})^3 \times (a_{B,STOP})^3 \\ & \quad \times (b_A(f))^1 \times (b_A(e))^2 \\ & \quad \times (b_B(h))^2 \times (b_B(g))^1 \end{aligned}$$

HMM - Supervised Learning (Example)

Let us use a simple labeled sequence dataset to help you understand how we estimate the parameters in the supervised learning setting.

- The labeled training dataset contains three sequence pairs:

$$\begin{array}{ccc} y^{(1)} = (A, B) & y^{(2)} = (A, B) & y^{(3)} = (A, B) \\ x^{(1)} = (e, g) & x^{(2)} = (e, h) & x^{(3)} = (f, h) \end{array}$$

$O = \{e, f, g, h\}$, and $I = \{START, A, B, STOP\}$

- 1) We first write down the likelihood (simplifying and abstracting it further!):

$$\begin{aligned} & (a_{START,A})^3 \times (a_{A,B})^3 \times (a_{B,STOP})^3 \\ & \times (b_A(f))^1 \times (b_A(e))^2 \\ & \times (b_B(h))^2 \times (b_B(g))^1 \end{aligned} \quad \Rightarrow \quad \begin{aligned} & (a_{START,A})^{\text{Count}(START,A)} \\ & \times (a_{A,B})^{\text{Count}(A,B)} \\ & \times (a_{B,STOP})^{\text{Count}(B,STOP)} \end{aligned}$$

HMM - Supervised Learning (Example)

Let us use a simple labeled sequence dataset to help you understand how we estimate the parameters in the supervised learning setting.

- The labeled training dataset contains three sequence pairs:

$$\begin{array}{lll} y^{(1)} = (A, B) & y^{(2)} = (A, B) & y^{(3)} = (A, B) \\ x^{(1)} = (e, g) & x^{(2)} = (e, h) & x^{(3)} = (f, h) \end{array}$$

$O = \{e, f, g, h\}$, and $I = \{START, A, B, STOP\}$

- 1) We first write down the likelihood (simplifying and abstracting it further!):

$$\begin{aligned} & (a_{START,A})^3 \times (a_{A,B})^3 \times (a_{B,STOP})^3 \\ & \times (b_A(f))^1 \times (b_A(e))^2 \\ & \times (b_B(h))^2 \times (b_B(g))^1 \end{aligned} \quad \Rightarrow \quad \begin{aligned} & (a_{START,A})^{\text{Count}(START,A)} \\ & \times (a_{A,B})^{\text{Count}(A,B)} \\ & \times (a_{B,STOP})^{\text{Count}(B,STOP)} \end{aligned} \quad \Rightarrow \quad \prod_{u,v} (a_{u,v})^{\text{Count}(u,v)}$$

HMM - Supervised Learning (Example)

Let us use a simple labeled sequence dataset to help you understand how we estimate the parameters in the supervised learning setting.

- The labeled training dataset contains three sequence pairs:

$y^{(1)} = (A, B)$	$y^{(2)} = (A, B)$	$y^{(3)} = (A, B)$
$x^{(1)} = (e, g)$	$x^{(2)} = (e, h)$	$x^{(3)} = (f, h)$

$O = \{e, f, g, h\}$, and $I = \{START, A, B, STOP\}$

- 1) Final general form of likelihood:

$$\prod_{u,v} (a_{u,v})^{\text{Count}(u,v)} \times \prod_{u,o} (b_u(o))^{\text{Count}(u \rightarrow o)}$$

HMM - Supervised Learning (Example)

Let us use a simple labeled sequence dataset to help you understand how we estimate the parameters in the supervised learning setting.

- The labeled training dataset contains three sequence pairs:

$y^{(1)} = (A, B)$	$y^{(2)} = (A, B)$	$y^{(3)} = (A, B)$
$x^{(1)} = (e, g)$	$x^{(2)} = (e, h)$	$x^{(3)} = (f, h)$

$O = \{e, f, g, h\}$, and $I = \{START, A, B, STOP\}$

- 2) Take the log-likelihood instead:

$$\sum_{u,v} \text{Count}(u, v) \log(a_{u,v}) + \sum_{u,o} \text{Count}(u \rightarrow o) \log(b_u(o))$$

HMM - Supervised Learning (Example)

Let us use a simple labeled sequence dataset to help you understand how we estimate the parameters in the supervised learning setting.

- The labeled training dataset contains three sequence pairs:

$y^{(1)} = (A, B)$	$y^{(2)} = (A, B)$	$y^{(3)} = (A, B)$
$x^{(1)} = (e, g)$	$x^{(2)} = (e, h)$	$x^{(3)} = (f, h)$

$O = \{e, f, g, h\}$, and $I = \{START, A, B, STOP\}$

3) Next, take the partial derivative with respect to each individual parameter, and set it to zero. Solve the equation.

$$a_{u,v} = \frac{\text{count}(u,v)}{\text{count}(u)} \quad b_u(o) = \frac{\text{count}(u \rightarrow o)}{\text{count}(u)}$$

Not tested but if interested, check out here:
<https://cs229.stanford.edu/section/cs229-hmm.pdf>
(see 1.1.2)



HMM - Supervised Learning (Summary)

- 1) We first write down the likelihood:

$$\prod_{u,v} (a_{u,v})^{\text{Count}(u,v)} \times \prod_{u,o} (b_u(o))^{\text{Count}(u \rightarrow o)}$$

- 2) We then take the log-likelihood:

$$\sum_{u,v} \text{Count}(u, v) \log(a_{u,v}) + \sum_{u,o} \text{Count}(u \rightarrow o) \log(b_u(o))$$

- 3) We take the partial derivative with respect to each individual parameter, and set it to zero. Solve the equation.

$$a_{u,v} = \frac{\text{count}(u,v)}{\text{count}(u)}$$

$$b_u(o) = \frac{\text{count}(u \rightarrow o)}{\text{count}(u)}$$

∴ it is too small for the joint P

Solve the partial derivative to find a & b.

What you should know?

- What is a **Hidden Markov Model (HMM)**? What **types of data** it can handle?
- What is the **relation** between **HMM** and **mixture model**?
- How to **describe the joint distribution** of a HMM with the **transition** and **emission probabilities**?
- How to **learn parameters** of HMM in **supervised** learning setting?

Suggestion: If you're not very clear or want to learn more, please do read: "C. Bishop: Pattern Recognition and Machine Learning. Springer, 2006" (recommended text book). It will help you to understand the concept.

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 - *Stanford University – Machine Learning (CS229)*